Dynamic Segment Trees : Online Queries for Range Sum with Point Updates

Last Updated : 12 Jul, 2025

Prerequisites: Segment Tree

Given a number **N** which represents the size of the array initialized to 0 and **Q** queries to process where there are two types of queries:

1. 1 P V: Put the value **V** at position **P**.
2. 2 L R: Output the sum of values from **L** to **R**.

The task is to answer these queries.

Constraints:

- 1 ? N ? 10^{18}
- Q ? 10^{5}
- 1 ? L ? R? N

Note: Queries are online. Therefore:

- L = (previousAnswer + L) % N + 1
- R = (previousAnswer + R) % N + 1

Examples:

> ***Input:*** *N = 5, Q = 5, arr[][] = {{1, 2, 3}, {1, 1, 4}, {1, 3, 5}, {1, 4, 7}, {2, 3, 4}}*
> ***Output:*** *12*
> ***Explanation:***
> *There are five queries. Since N = 5, therefore, initially, the array is {0, 0, 0, 0, 0}*

For query 1: 1 2 3 array = {0, 3, 0, 0, 0}
For query 2: 1 1 4 array = {4, 3, 0, 0, 0}
For query 3: 1 3 5 array = {4, 3, 5, 0, 0}
For query 4: 1 4 7 array = {4, 3, 5, 7, 0}
For query 5: 2 3 4 Sum from [3, 4] = 7 + 5 = 12.

Input: *N = 3, Q = 2, arr[][] = {{1, 1, 1}, {1, 2, 2}, {1, 3, 3}}*
Output: *0*

Approach: Here, since the updates are high, Kadane's algorithm doesn't work quite well. Moreover, since it is given that the queries are online, a simple segment tree would not be able to solve this problem because the constraints for the number of elements is very high. Therefore, a new type of data structure, a dynamic segment tree is used in this problem.
Dynamic Segment Tree: Dynamic segment tree is not a new data structure. It is very similar to the segment tree. The following are the properties of the dynamic segment tree:

- Instead of using an array to represent the intervals, a node is created whenever a new interval is to be updated.
- The following is the structure of the node of the dynamic segment tree:

C++ Java Python3 C# JavaScript

```
// Every node contains the value and
// the left subtree and right subtree
struct Node {
    long long value;
    struct Node *L, *R;
};

struct Node* getnode()
```

```
{
    struct Node* temp = new struct Node;
    temp->value = 0;
    temp->L = NULL;
    temp->R = NULL;
    return temp;
}
```

- Clearly, the above structure is the same as a Binary Search Tree. In every node, we are storing the node's value and two pointers pointing to the left and right subtree.
- The Interval of the root is from [1, N], the interval of the left subtree will be [1, N/2] and the interval for the right subtree will be [N/2 + 1, N].
- Similarly, for every node, we can calculate the interval it is representing. Let's say the interval of the current node is [L, R]. Then, the Interval of its left and right subtree are [L, (L + R)/2] and [(L + R)/2+1, R] respectively.
- Since we are creating a new node only when required, the build() function from the segment tree is completely removed.

Before getting into the algorithm for the operations, let's define the terms used in this article:

- **Node's interval:** It is the interval the node is representing.
- **Required interval:** Interval for which the sum is to calculate.
- **Required index:** Index at which Update is required.

The following is the algorithm used for the operations on the tree with the above-mentioned properties:

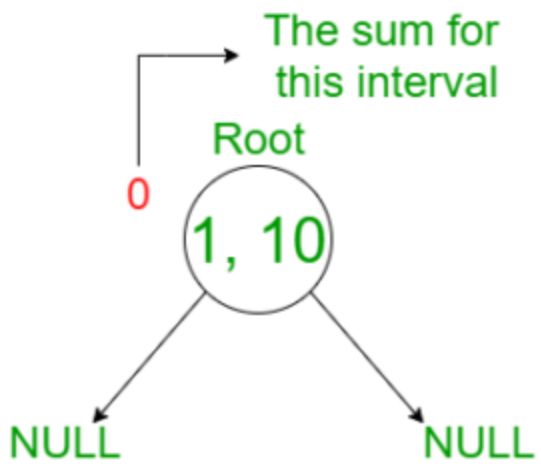
1. **Point Update:** The following algorithm is used for the point update:
    1. Start with the root node.
    2. If the interval at the node doesn't overlap with the required index then return.

    3. If the node is a NULL entry then create a new node with the appropriate intervals and descend into that node by going back to step 2 for every new child created.
    4. If both, intervals and the index at which the value is to be stored are equal, then store the value into at that node.
    5. If the interval at the node overlaps partially with the required index then descend into its children and continue the execution from step 2.
2. **Finding the sum for every query:** The following algorithm is used to find the sum for every query:
    1. Start with the root node.
    2. If the node is a NULL or the interval at that node doesn't overlap with the required interval, then return 0.
    3. If the interval at the node completely overlaps with the required interval then return the value stored at the node.
    4. If the interval at the node overlaps partially with the required interval then descend into its children and continue the execution from step 2 for both of its children.

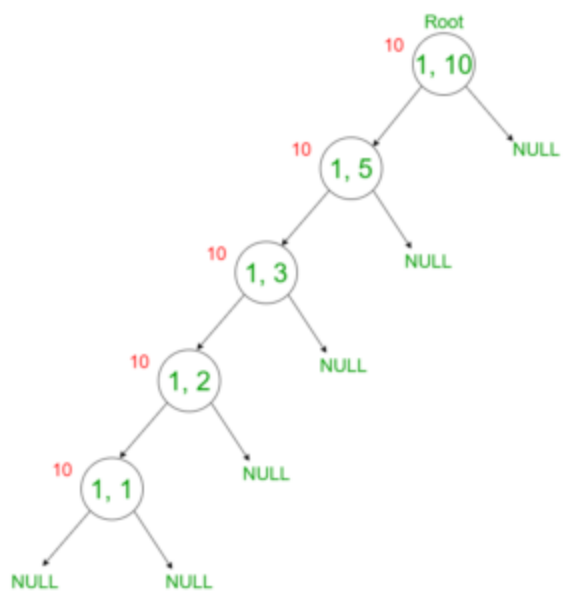
Example: Lets visualize the update and sum with an example. Let N = 10 and the operations needed to perform on the tree are as follows:

1. Insert 10 at position 1.
2. Find the sum of value of indices from 2 to 8.
3. Insert 3 at position 5.
4. Find the sum of value of indices from 3 to 6.

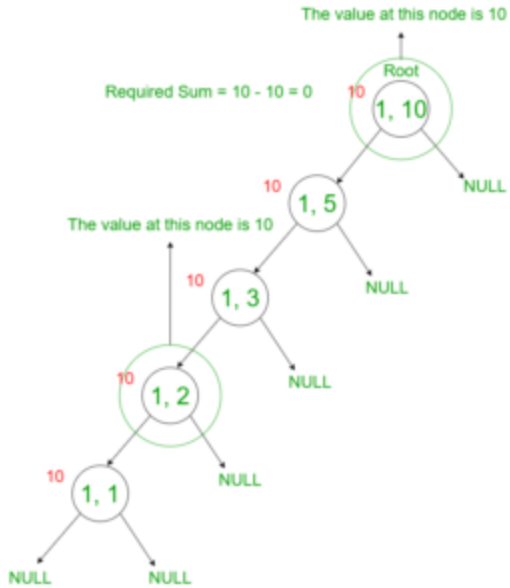
- Initially, for the value N = 10, the tree is empty. Therefore:



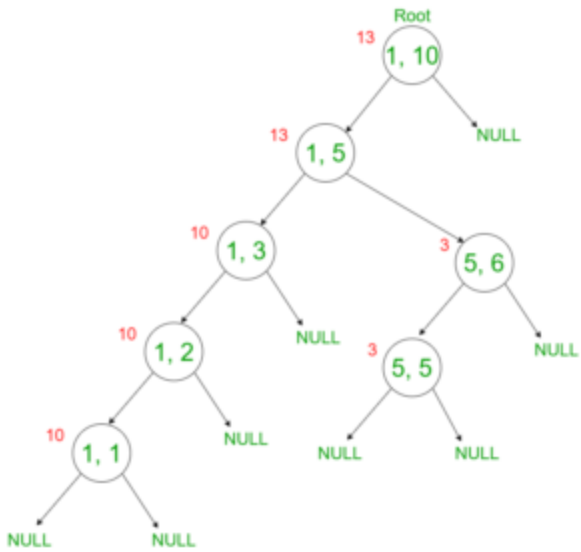
- Insert 10 at position 1. In order to do this, create a new node until we get the required interval. Therefore:



- Find the sum of value of indices from 2 to 8. In order to do this, the sum from [1, 8] is found and the value [1, 2] is subtracted from it. Since the node [1, 8] is not yet created, the value of [1, 8] is the value of the root [1, 10]. Therefore:



- Insert 3 at position 5. In order to do this, create a new node until we get the required interval. Therefore:



Below is the implementation of the above approach:

C++ Java Python3 C# JavaScript

```
// C++ program for the implementation
// of the Dynamic segment tree and
// perform the range updates on the
// given queries
```

```
#include <bits/stdc++.h>

using namespace std;
typedef long long ll;

// Structure of the node
struct Node {

    ll value;
    struct Node *L, *R;
};

// Structure to get the newly formed
// node
struct Node* getnode()
{
    struct Node* temp = new struct Node;
    temp->value = 0;
    temp->L = NULL;
    temp->R = NULL;
    return temp;
```



```
// Creating the Root node
struct Node* root;

// Function to perform the point update
// on the dynamic segment tree
void UpdateHelper(struct Node* curr, ll index,
                  ll L, ll R, ll val)
{

    // If the index is not overlapping
    // with the index
    if (L > index || R < index)
        return;

    // If the index is completely overlapping
    // with the index
    if (L == R && L == index) {
```

```
        // Update the value of the node
        // to the given value
        curr->value = val;
        return;
    }

    // Computing the middle index if none
    // of the above base cases are satisfied
    ll mid = L - (L - R) / 2;
    ll sum1 = 0, sum2 = 0;

    // If the index is in the left subtree
    if (index <= mid) {

        // Create a new node if the left
        // subtree is null
        if (curr->L == NULL)
            curr->L = getnode();

        // Recursively call the function
        // for the left subtree
        UpdateHelper(curr->L, index, L, mid, val);
    }

    // If the index is in the right subtree
    else {

        // Create a new node if the right
        // subtree is null
        if (curr->R == NULL)
            curr->R = getnode();

        // Recursively call the function
        // for the right subtree
        UpdateHelper(curr->R, index, mid + 1, R, val);
    }

    // Storing the sum of the left subtree
    if (curr->L)
```

```
        sum1 = curr->L->value;

    // Storing the sum of the right subtree
    if (curr->R)
        sum2 = curr->R->value;

    // Storing the sum of the children into
    // the node's value
    curr->value = sum1 + sum2;
    return;
}

// Function to find the sum of the
// values given by the range
ll queryHelper(struct Node* curr, ll a,
               ll b, ll L, ll R)
{

    // Return 0 if the root is null
    if (curr == NULL)
        return 0;

    // If the index is not overlapping
    // with the index, then the node
    // is not created. So sum is 0
    if (L > b || R < a)
        return 0;

    // If the index is completely overlapping
    // with the index, return the node's value
    if (L >= a && R <= b)
        return curr->value;

    ll mid = L - (L - R) / 2;

    // Return the sum of values stored
    // at the node's children
    return queryHelper(curr->L, a, b, L, mid)
           + queryHelper(curr->R, a, b, mid + 1, R);
}
```

```
// Function to call the queryHelper
// function to find the sum for
// the query
ll query(int L, int R)
{
    return queryHelper(root, L, R, 1, 10);
}

// Function to call the UpdateHelper
// function for the point update
void update(int index, int value)
{
    UpdateHelper(root, index, 1, 10, value);
}

// Function to perform the operations
// on the tree
void operations()
{
    // Creating an empty tree
    root = getnode();

    // Update the value at position 1 to 10
    update(1, 10);

    // Update the value at position 3 to 5
    update(3, 5);

    // Finding sum for the range [2, 8]
    cout << query(2, 8) << endl;

    // Finding sum for the range [1, 10]
    cout << query(1, 10) << endl;

}

// Driver code
int main()
{
```

```
    operations();

    return 0;
}
```

Output:

```
5
15
```

Time Complexity: O(Q * logN)
Auxiliary Space: O(N)

Related Topic: Segment Tree

Comment More info

Explore

DSA Fundamentals

Data Structures

Algorithms

Advanced

Interview Preparation

Practice Problem

GeeksforGeeks
Sanchhaya Education Private Limited

Corporate & Communications Address:

A-143, 7th Floor, Sovereign Corporate Tower, Sector- 136, Noida, Uttar Pradesh (201305)

Registered Address:

K 061, Tower K, Gulshan Vivante Apartment, Sector 137, Noida, Gautam Buddh Nagar, Uttar Pradesh, 201305

GET IT ON Google Play
Download on the App Store

Company

About Us
Legal
Privacy Policy
Contact Us
Advertise with us
GFG Corporate Solution
Campus Training Program

Explore

POTD
Job-A-Thon
Community
Blogs
Nation Skill Up

Tutorials

Programming Languages
DSA
Web Technology
AI, ML & Data Science
DevOps
CS Core Subjects
Interview Preparation
GATE
Software and Tools

Courses

IBM Certification
DSA and Placements
Web Development
Programming Languages
DevOps & Cloud
GATE
Trending Technologies

@GeeksforGeeks, Sanchhaya Education Private Limited, All rights reserved